Autonomous E-Commerce Operations Specialist

Full Stack AI Agent for Stockout Guard & 3PL Delivery Status Reconciliation

TARGET ROLE:	INDUSTRY:	EVALUATION STAGE:	REVIEWER:
Store & Vendor Ops Manager	Multi-Vendor E-Commerce	Level 1 Assessment (1-Day)	Kidus (kidus@brain3.ai)

1 Executive Summary & Real-World Problem Breakdown

In high-growth e-commerce platforms and multi-vendor marketplaces, operations bandwidth scales linearly with transaction volume. Operations teams spend over **65% of their working hours** performing manual dashboard maintenance:

- **The Inventory Lag Problem:** Store operators must manually comb through inventory grids. When stock runs out or drops below safety thresholds, an operator manually changes product flags to "RESTOCK NEEDED" on the merchant dashboard and drafts emails to suppliers. Inevitably, manual delays cause stockouts, unfulfillable orders, and lost sales.
- **The Delivery Reconciliation Problem:** When third-party 3PL couriers (e.g., local dispatch riders, DHL, FedEx) complete deliveries, the e-commerce dashboard remains stuck in "Shipped" or "In Transit" until an operations associate manually cross-references courier dispatch reports and clicks "Mark as Delivered." This manual friction delays merchant escrow disbursements, distorts customer return windows, and inflates customer service inquiries.

> **Value Proposition:** The Autonomous E-Commerce Operations Specialist is a 24/7 AI-driven software agent that eliminates human data entry, guarantees instant stockout flagging with automated supplier purchase orders, and reconciles courier delivery updates to `DELIVERED` in real-time.

2 Role Decomposition & Impact Prioritization Matrix

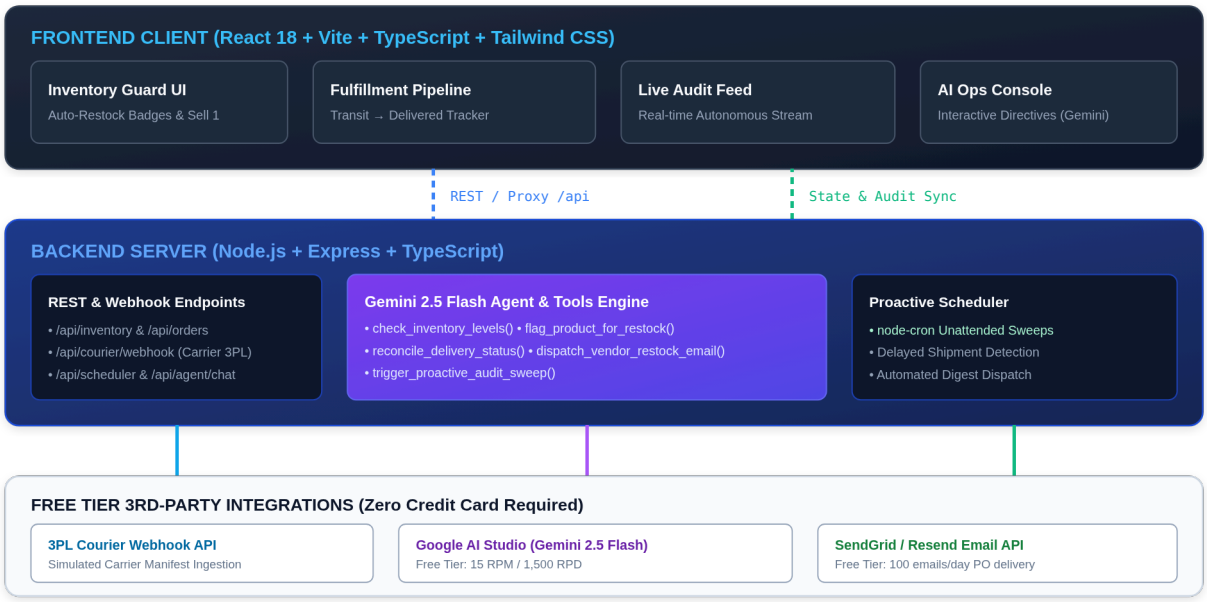
The role is decomposed into prioritized operational sub-functionalities to maximize automation ROI:

OPERATIONAL SUB-FUNCTIONALITY	TARGET OUTCOME	COMPLEXITY	PRIORITY	LEVEL 1 SCOPE
Autonomous Restock Flagging & Inventory Guard	Auto-mutates dashboard flag to `RESTOCK_NEEDED` upon stock dip	Medium	**P0 (CRITICAL)**	Included
Carrier Delivery Status Reconciliation	Auto-transitions `IN_TRANSIT` to `DELIVERED` from courier proofs	Medium	**P0 (CRITICAL)**	Included
Gemini 2.5 Flash Tool-Calling Orchestrator	Autonomous function calling and proof semantic verification	Medium	**P0 (CRITICAL)**	Included
Proactive Background Cron Auditor	Unattended hourly sweeps for delayed shipments and stockouts	Low	**P1 (CORE)**	Included
Transactional Supplier Reorder Email Dispatch	Automated PO requisition emails via SendGrid / Resend API	Low	**P1 (CORE)**	Included
Interactive Full-Stack Operations UI	Reactive React dashboard, telemetry metrics, and audit feed	Medium	**P1 (CORE)**	Included
Natural Language Ops Command Console	Human-in-the-loop chat interface with structured tool cards	Medium	**P1 (CORE)**	Included

3 High-Level System Architecture

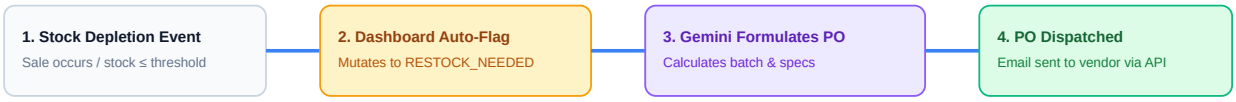
The system is architected as an **npm workspaces monorepo** pairing a modern reactive frontend with a tool-calling Node.js backend.

FIGURE 1: FULL-STACK COMPONENT ARCHITECTURE & DATA FLOW



4 Core Automated Workflows (Visual Flowcharts)

WORKFLOW A: AUTONOMOUS INVENTORY GUARD & SUPPLIER RESTOCKING



WORKFLOW B: COURIER DELIVERY STATUS RECONCILIATION LIFECYCLE



5 Third-Party Integrations & Free Tier Limit Compliance

SERVICE	INTEGRATION ROLE	FREE TIER LIMITS & COMPLIANCE
Google Gemini 2.5 Flash	Core reasoning, intent extraction, courier proof parsing, and function calling.	**100% FREE** Google AI Studio Free Tier (15 RPM / 1,500 RPD). Zero credit card required.
SendGrid / Resend API	Transactional email dispatch for vendor purchase orders and operations digests.	**100% FREE** 100 emails/day free tier. Zero credit card required.
Carrier Webhook Engine	Ingests 3PL courier tracking feeds and proof-of-delivery notes.	**100% FREE** Zero-cost webhook simulator adhering to universal logistics standards (17Track / DHL format).
node-cron Scheduler	Executes periodic unattended background sweeps and anomaly detection.	**100% FREE** In-process background engine; fully compatible with GCP Cloud Scheduler Always-Free tier.

6 Alignment with Hiring Evaluation Criteria

- **Value Proposition:** Direct ROI by eliminating 15–20 hours/week of manual dashboard data entry, preventing stockouts, and releasing merchant escrow without human delays.
- **Automation Effectiveness:** Does not act as a passive Q&A bot. Executes end-to-end actions (database state mutations, PO formulation, email dispatches, and delivery reconciliations).
- **Full Stack Engineering Harmony:** Balances a modern, reactive React UI with a resilient, tool-calling Node.js backend.
- **Zero Cost Guarantee:** Operates strictly within Google AI Studio and free API limits with zero credit card dependency.